

natex 0.3.0: Methods for Automated Natural-Experiment Discovery and Calibrated Inference in Tabular Data

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methods of record for **natex** v0.3.0, 2026-10-05

Abstract

natex is an open-source toolkit that searches tabular data for natural experiments and estimates their effects. This paper is the methods record for the paper collection built with it. One command runs seven design families: local regression-discontinuity discovery (LoRD3), subset difference-in-differences discovery (SuDDDS), known-cutoff regression kink and difference-in-kinks designs, Lasso instrument search with Anderson–Rubin inference, synthetic-control donor selection, bunching at declared thresholds, and debiased effect estimation. Discovery never reads the outcome. Version 0.3.0 enumerates the full role space of treatments, running variables and panels. It no longer scans only the configurations a heuristic or a language model proposed. Discovered designs are calibrated by fitted-null Monte Carlo with an honest discovery/estimation split. Calendar-time kinks are gated on a placebo-calibrated p over at least 19 shifted cutoffs. Designs that cannot be validly tested receive an *inconclusive* verdict rather than a null. On six public benchmark datasets the pipeline recovers every published design blind. On eight AI time series the nominal 5% kink test rejected at 7% to 88% of placebo dates. That is why the nominal p no longer decides any verdict.

1 Introduction

Natural experiments in observational data are usually found by hand. A researcher knows a rule, locates its cutoff and checks whether the design holds. Herlands et al. [14] proposed to automate the search. Fit a smooth model of treatment assignment, scan local neighbourhoods for a boundary where assignment jumps, and validate each boundary. **natex** [10] implements that programme together with the subset scan for difference-in-differences from Herlands’s thesis [15] and the debiasing layer of Jakubowski et al. [17]. It adds known-cutoff regression kink and difference-in-kinks designs [5, 4], Lasso instrument search [3], synthetic-control donor selection [1] and bunching tests at declared thresholds [18]. It is a Python package with a command-line interface and a Python API.

This paper records the methods as they stand in version 0.3.0. It states each estimand, the inference of record, and the checks behind every verdict. It then gives two kinds of evidence. Six backtests show what the procedures recover when the answer is published. Eight AI time series show how often the nominal kink test rejects at dates where nothing happened. Two worked examples

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from October 2026 illustrate the full pipeline on fresh data. The empirical papers in the collection take their inference rules from here.

Three changes in 0.3.0 matter for reading the collection. First, the scan enumerates every treatment, running-variable and panel combination that the data profile admits. An analyst pass, heuristic or language model, orders the scan but no longer bounds it. Second, a design the pipeline could not validly test is reported as inconclusive. Under 0.2.0 a refused placebo test, an unscannable configuration and a degenerate fit were all reported as null. Third, a kink in calendar time is judged by its rank among shifted placebo cutoffs, not by its heteroskedasticity-robust p -value. Section 7.2 shows why.

Section 2 gives the pipeline and the design families. Section 3 states the discovery statistics. Section 4 states the inference of record. Section 5 lists the corrections to the source papers that the implementation carries. Section 6 defines the verdict vocabulary. Section 7 reports the evidence. Section 8 states what the methods cannot do.

2 Pipeline and design families

A run starts from a CSV file. The intake profiler classifies every column as numeric, binary, categorical or time-like. It lists panel candidates: unit and time column pairs whose grid covers at least 95% of rows. Elapsed-time names such as `days`, `days_since_x`, `week`, `period`, `wave` and `t` count as time-like. An optional analyst pass proposes column roles, a declarative preparation plan and a ranked search plan. The default analyst is a deterministic heuristic that needs no network. Language-model backends are available but advisory: the same data and seed give bitwise-identical statistics with and without guidance.

The survey command then runs the seven families of Table 1 in a fixed order and writes one report with a verdict per family. Each family runs in isolation. An exception is recorded verbatim and the survey continues. A single `numpy.random.Generator` is spawned once into seven streams in registry order, so a skipped family never shifts another family’s random draws. Same seed, same report.

The house rules apply to every family. Discovery never reads the outcome. A failed computation returns a missing value, never zero. Every stochastic step draws from the one generator. Where a source paper and its released code disagree with the derivation, the reconciled audit of Section 5 governs.

3 Discovery statistics

3.1 LoRD3: local regression-discontinuity discovery

Let x_i be the forcing columns, T_i the treatment and y_i the outcome. LoRD3 fits one smooth background model $T_i = f(x_i) + \varepsilon_i$, with f a polynomial of degree d in the standardised forcing columns. The residual model is heteroskedastic Normal for a real-valued treatment and Bernoulli with a log-odds offset for a binary one. The fit reads only (x, T) .

The scan visits every data point. The neighbourhood of point i is its k nearest neighbours in standardised forcing space. Each of the $k - 1$ hyperplanes through the centre and one other neighbour bisects the neighbourhood. A bisection is scored by the log-likelihood ratio of a two-group shift alternative against a single-shift null. For the Normal model the score is the generalised likelihood ratio of two side means against one common mean. The weights are $1/\sigma_j^2$ for residual variances σ_j^2 . For the Bernoulli model the side offsets are fitted by bracketed Newton steps in $\log \beta$

Table 1: Design families. “Discovered” means the cutoff, subset or instrument set is searched for; “declared” means the user supplies it.

Family	Inputs	Cutoff	Statistic	Output
rdd	treatment, numeric forcing columns	discovered	LoRD3 max LLR over local hyperplane splits	local effect at the boundary, frozen-side 2SLS
did	unit $\times$ time panel, treatment variable	discovered	SuDDDS LLR over covariate subsets and intervention times	DD, synthetic-control or GESS gap with placebo p
kink	outcome, running variable, cutoff, bandwidth	declared	right-minus-left local-linear slope change	RKD or DiK marginal response; descriptive slope change in calendar time
iv	treatment, instrument pool, controls	discovered	plug-in Lasso first stage, HC1 F , Hansen J	2SLS with Anderson–Rubin set on the estimation half
sc	unit $\times$ time outcomes, treated unit, t_0	declared	pre-period fit, post/pre RMSPE ratio	post-period ATT with in-space placebo p
bunching	running variable, thresholds	declared	binned-Poisson intercept jump	density jump at the threshold
dee	covariates, treatment, outcome, observational CATE estimator	discovered via rdd	GP bias surface, stacked mixture	debiased CATE surface

with a Firth fallback under separation. A pure-group split, where one side is all treated, is scored at the boundary likelihood supremum rather than dropped. Discoveries are ranked by maximum LLR. The responsible forcing columns are read from the normalised hyperplane normal. The centre lies on every candidate hyperplane, so membership is explicit: signed distance at least zero is group one.

Defaults are $k = 50$, degree $d = 1$ and $q = 99$ null replicas. For large n a coarse-to-fine scan scores a seeded subsample of 2,000 centres and rescans the top candidates at full resolution. The share of centres scanned is always reported. On the 44,362-row academic-probation file the coarse-to-fine scan takes about five seconds.

3.2 The role space

A configuration is a treatment column, a set of forcing columns and, for did, a unit and time pair. Version 0.3.0 enumerates every profiled binary treatment candidate crossed with every single non-time-like forcing column. It adds the joint all-forcing candidate per treatment and every treatment crossed with every profiled panel. Three kinds of column are excluded, each with its reason recorded. They are time-like forcing columns, columns declared as outcomes in the preparation plan, and treatments that are a global deterministic step in a time column. The last case is a mechanical rediscovery of the design’s own adoption boundary and is reported as inconclusive whatever its scan p .

A search plan’s guessed outcome is not reserved. It stays in the forcing space and is estimated as an outcome wherever it is not a role. Under 0.2.0 the heuristic reserved the first numeric column, which on a test fixture removed the planted running variable from every scan. Every scanned boundary is now estimated against every candidate outcome, and the report carries a per-outcome table. The scan still never reads any of them. Configurations are ranked by fitted-null p before raw LLR, because LLRs are not comparable across forcing dimensions. The report lists what was

Table 2: Kink estimands. $\kappa[\cdot]$ is the right-minus-left slope change of the named variable; “policy” is the observed policy variable in fuzzy designs.

Design	Estimand
Sharp RKD	$\kappa[Y]/\text{known policy kink}$
Fuzzy RKD	$\kappa[Y]/\kappa[\text{policy}]$
Sharp DiK	$(\kappa[Y, \text{post}] - \kappa[Y, \text{pre}])/\text{known policy kink change}$
Fuzzy DiK	$(\kappa[Y, \text{post}] - \kappa[Y, \text{pre}])/(\kappa[\text{policy}, \text{post}] - \kappa[\text{policy}, \text{pre}])$
Group DiK	as DiK with group 1 and group 0 in place of post and pre

enumerated, what was excluded, and what the budget left unscanned.

3.3 SuDDDS: subset difference-in-differences discovery

Panel records $(x_i, t_i, \theta_i, y_i)$ carry categorical covariates x , time t , a treatment variable θ and an outcome y . The background model gives every unit its own effect plus a polynomial in standardised time, solved by least squares. The residuals $r_i = \theta_i - \hat{f}(\text{unit}_i, t_i)$ have profile-level variances σ_i^2 shrunk towards the global variance with a data-scaled floor. The scan looks for a subset s and a time T_0 at which θ jumps inside a window $[T_0 - W, T_0 + W)$. A subset is a conjunction over dimensions of unions over values. With $c_i = r_i/\sigma_i^2$, $b_i = 1/\sigma_i^2$ and C_1, B_1 the post-side sums and C_2, B_2 the pre-side sums over s , the double-beta score is

$$\text{LLR}(s, T_0) = \frac{C_1^2}{2B_1} + \frac{C_2^2}{2B_2} - \frac{(C_1 + C_2)^2}{2(B_1 + B_2)}. \quad (1)$$

The corrected single- Δ statistic profiles the per-profile means out under both hypotheses and restricts to the window. For a binary θ the window contrast is scored with exact Bernoulli log-odds offsets; the Normal model remains available for parity with the thesis. A global incumbent is kept across all windows and restarts. A cutoff qualifies only with at least three in-window records on each side. Dimensions with at most 12 values are enumerated exactly, $2^k - 1$ subsets per dimension. Default windows are one eighth, one quarter and one half of the time span, with eight restarts.

3.4 Known-cutoff kinks

The kink estimator fits a separate local polynomial on each side within $|x| \leq h$ of a declared cutoff. Triangular kernel weights and degree one are the defaults. A kink is always right minus left: $\kappa[Z] = m[Z, \text{right}] - m[Z, \text{left}]$, with m the one-sided derivative of $E[Z \mid x]$ at zero. Table 2 gives the estimands. A group difference-in-kinks contrasts a treated group’s kink against a control group’s at the same cutoff. It is numerically the stacked DiK estimator with the group indicator in place of the period indicator. What changes is the identifying assumption, which becomes parallel non-policy kinks across groups.

With a unit policy denominator the estimate is a descriptive slope change in outcome units per unit of the running variable. That is the form every calendar-time kink in the collection takes. The bandwidth is required. The survey’s default, the median absolute distance to the cutoff, carries no optimality claim. It is reported with a half, one and double bandwidth grid.

3.5 Instruments, donors and bunching

Instrument search follows Belloni et al. [3]. The treatment and the candidate pool are residualised on the controls. An iterated plug-in Lasso with heteroskedasticity-robust loadings selects a support;

the penalty is $\lambda = 2c\sqrt{n}\Phi^{-1}(1 - \gamma/2p)$ with $c = 1.1$ and $\gamma = 0.1/\log \max(n, p)$. Post-Lasso OLS gives the first stage. An empty selection is reported as empty, with a missing F . Selection reads only the treatment, the pool and the controls.

Donor selection follows Abadie, Diamond and Hainmueller [1]. Candidates are units observed at every pre-period time of the treated unit. They are scored on the pre-period trajectory only, and simplex weights are fitted on a scale-normalised objective. The post-period gap gives the ATT. No covariate weights are fitted.

Bunching at a declared threshold reuses the density statistic of Section 4.2. A binned-Poisson regression of counts on signed distance, with a bin edge pinned at the threshold, tests the intercept jump. Because the threshold was declared, no selection correction is needed.

4 Inference of record

4.1 Fitted-null Monte Carlo and the honest split

The significance of a discovered boundary is a parametric bootstrap against the fitted null. Each replica redraws T^* from the fitted background model and reruns the full scan. Binary replicas are drawn as $\text{Bernoulli}(\hat{p})$ directly. The p -value is the add-one rank,

$$p = \frac{1 + \#\{\text{replica max LLR} \geq \text{observed max LLR}\}}{q + 1}. \quad (2)$$

This is never called exact. The primary post-selection guarantee is an honest split: a random half of the rows is used for discovery and the disjoint half for estimation. The selection event is then a function of the discovery half alone, so estimation-half p -values keep their nominal distributions without correction. The same split separates instrument selection from 2SLS, J and Anderson–Rubin inference.

4.2 Falsification battery for discovered designs

Every discovered boundary passes through three checks. Placebo tests fit side-specific trends in each remaining covariate and test intercept equality at the cutoff, with Holm correction across covariates [16]. The printed side-mean contrasts were dropped because any smooth covariate rejects them under a valid design. A density test regresses binned counts of the signed distance along the frozen discovered normal on a polynomial with an intercept jump, in the spirit of McCrary [19]. It is a falsification check only, because the search selected the geometry. First-stage relevance is never inferred from a high LLR. The first-stage jump, its t -statistic, a weak-instrument flag and an Anderson–Rubin set [2, 13] accompany every local 2SLS estimate.

SuDDDS discoveries are checked for composition and anticipation. The composition check is a chi-square test of unit counts before and after T_0 inside the window. The anticipation check tests placebo jumps at T_0 minus one to three steps, with Holm correction. A McCrary test on calendar time is information-free when record times are design-determined, and is not run.

4.3 Calendar-time kinks: placebo calibration

A kink in calendar time asks whether a trend bent at a dated event. No unit sorts itself across a date, so density arguments give no protection. Identification reduces to one untestable assumption: nothing else bent the outcome’s expected slope at that date. Two facts make the nominal p unusable as the gate. Aggregate series are serially correlated, and side-specific local fits absorb low-frequency

Table 3: Empirical size at nominal 5% of the kink t -test on 64 time points of AR(1) noise with $\rho = 0.75$ and no kink.

Covariance	Empirical size
HC1	0.48
Newey–West, lags 4 to 20	0.34
VAR(1) prewhitening	0.19

noise before the residual scores see it. Table 3 gives the calibration on AR(1) noise. At nominal 5% the HC1 test rejects about half the time. A Newey–West covariance still rejects a third of the time at every lag between 4 and 20.

The inference of record is therefore a placebo-calibrated p . The same estimator, with the same bandwidth, kernel and covariance, is run at shifted placebo cutoffs. The grid takes the distinct running values in the central 80% of the support. It drops any within a quarter bandwidth of the declared cutoff and thins to at most 49 positions (Appendix A). With N evaluable placebos and $z = \hat{\tau}/\widehat{se}$,

$$p_{\text{cal}} = \frac{1 + \#\{|z_{\text{placebo}}| \geq |z|\}}{1 + N}. \quad (3)$$

The smallest attainable value is $1/(N + 1)$. Fewer than 19 positions cannot reach 0.05, and the cutoff is reported as inconclusive. A seven-placebo grid has a floor of 0.125, so “0 of 7 placebos reject” is the floor of the test, not a 5% finding. Under 0.2.0 several collection papers made exactly that claim.

The grid yields a second number: the share of placebo cutoffs at which the nominal test rejects at 5%. This is the empirical size of the nominal test on that series, at dates where nothing happened. It is reported beside every verdict as a diagnostic. It is not a power certificate: rejections elsewhere say nothing about whether a bend at the event date would have been detected. Power is argued from the minimum detectable kink, about 1.96 standard errors against the slope change the claim implies.

The grid also separates bend existence from date attribution. A significant kink at the declared date together with significant kinks at shifted dates means the series bends over an era, not at the event. Two further safeguards apply. Cells with at most $d + 2$ points raise a small-cell warning. Leverage near one shrinks squared residuals and overstates HC1 precision. On a 12-point series the DiK z had $P(|z| \geq 3.63) = 0.080$ under its own fitted null, against a nominal 3.6×10^{-4} . And every estimate reports the lag-one autocorrelation of per-date mean residuals. A warning fires at 0.25 when no HAC covariance was requested.

4.4 Effects after discovery

A SuDDDS effect is tested with a two-sided studentised placebo statistic $|\tau_p/se_p|$. The placebo subsets are matched in shape to the discovered one, and the add-one rank is the p -value. Fewer than five usable placebos is a refusal and the family verdict is inconclusive. A synthetic-control effect is tested in space. Every complete donor is refit as pseudo-treated under the identical selection rule, with the treated unit removed from every placebo panel. The statistic is the post/pre RMSPE ratio. The gate for this family is 0.10. The add-one p has granularity $1/(n + 1)$, so 0.05 is often unattainable with few donors. The p itself is always reported. Instrument estimates carry a closed-form Anderson–Rubin set that is reported as found: interval, disjoint, unbounded or empty. It is never coerced to a finite interval, and it is bounded exactly when the homoskedastic first-stage F

Table 4: Design-level corrections carried by the implementation.

As printed	In <code>natex</code>
Randomization test described as exact	Parametric bootstrap against the fitted null, add-one rank; honest split as the post-selection guarantee
Bernoulli null drawn as $1\{\hat{p} + \hat{\sigma}Z > U\}$, success probability not $\hat{p}$	Replicas drawn as Bernoulli($\hat{p}$)
Placebo tests on side means, which reject any smooth covariate	Local intercept-continuity contrasts with side-specific trends, Holm-corrected
Group-instrument estimator $W = T - \mu$, inconsistent under the paper’s DGP	Frozen side-indicator 2SLS, HC1; the printed form is never implemented
One-sided 95th-percentile $\hat{\tau}$ placebo rule	Two-sided studentised statistic, matched placebo shapes, add-one p
Density test on an oblique projection with fixed critical values	Binned-Poisson test on the frozen discovered normal, falsification only
Scan LLR taken to imply instrument relevance	First-stage jump, t , weak flag and AR set reported with every estimate
DEE mixture draws a model label per prediction point	One label per posterior draw; analytic mixture covariance with the between-model term
Observational estimator fit on the rows that produced the local effects	Experiment-level cross-fitting; chi-square measurement model on SE^2
SuDDDS Algorithm 6 keeps no global best; Algorithm 7 scores empty sides	Global incumbent across windows and restarts; three records per side required
Algorithm 8 is deletion-only; Algorithm 9 maximises the control MSE	Relax-and-slice repair with exact enumeration up to 12 values; argmin with $+\infty$ start
Single- Δ statistic treats profile means as known	Means profiled out under both hypotheses; window-restricted; both signs scanned
Variance floor of 10^{-6} , absolute	Data-scaled floor; the released code’s reverse-neighbour variance indexing is not reproduced

exceeds the AR critical value.

5 Corrections to the source papers

Before implementation, all 142 numbered equations and algorithms of the three source papers were re-derived by one model family. A second model family re-derived them independently. Disagreements were settled by rendering the disputed pages at 576 dpi. The reconciled audit lives in the repository and governs whenever code, documentation and the printed papers disagree. Table 4 lists the corrections with design consequences. Typographical errors in the printed equations are absorbed silently and listed in the method cards.

6 Verdicts

Every family ends in one of six statuses (Table 5). The distinction that matters most is between null and inconclusive. Null means the design was testable and nothing credible surfaced. Inconclusive means the family ran but the test could not be valid. There are five reasons. A placebo pool has fewer than five usable members. A configuration could not be scanned. A treatment is a mechanical rediscovery of a constructed time step. A kink, density or DEE fit is degenerate. A calendar-time kink has fewer than 19 placebo positions. An inconclusive verdict is never a negative finding and

Table 5: Family statuses in the survey report.

Status	Meaning
credible	the family ran and its design-specific gate passed
null	the family ran, the design was testable, and no credible design surfaced
inconclusive	the family ran but the design could not be validly tested
skipped	the family did not run: inapplicable by profile, analyst skip, or a run-time gate
needs_input	a requirement only the user can supply is missing, such as a cutoff or an instrument pool
failed	the family raised; the error is recorded verbatim and other families are unaffected

never appears in an abstract.

The credible gates, with $\alpha = 0.05$, are:

- rdd: scan $p \leq \alpha$, placebo battery passed, density $p > \alpha$.
- did: scan $p \leq \alpha$; composition and anticipation checks passed.
- kink: minimum Holm-adjusted p across the declared cutoffs at or below α . The per-cutoff p is the nominal HC1 p for a cross-sectional running variable and the placebo-calibrated p for a time-like one.
- iv: instruments selected, with a strong first stage on both halves of the honest split.
- sc: in-space placebo p at or below 0.10.
- bunching: minimum Holm-adjusted binned-Poisson p at or below α .
- dee: a fitted surface over a usable ensemble.

Applicability is decided by declarative predicates over the profile and the user’s declarations. The data values are never read. An analyst may override the heuristic in either direction. Every override is recorded with its reason.

7 Evidence

7.1 Backtests on published designs

Six public datasets with a published design are run blind: the pipeline is never told the cutoff, the treated unit or the timing. Table 6 summarises what it recovers. The runs are the repository’s backtest suite, 32 tests in all, and the Prop 99 case has its own note [9].

Two Prop 99 details are findings about the estimators rather than the scan. With the corrected two-sided studentised placebo test, the three effect estimates rank 13, 7 and 9 of 39 placebo states. The thesis’s claim that all three are significant at 5% does not survive. And the in-space placebo rank of 3 of 39 differs from the published 1 of 39. The published figure discards poorly fitting placebos before ranking. That exclusion is opt-in here.

Table 6: Blind backtests. Row counts are after the documented protocol restrictions.

Dataset	Design	Recovered
MDRC RDD practice set, 2,767 rows	sharp RDD	pretest cutoff at 215; the known effect of about 10 lies inside the 2SLS interval
Academic probation (Lindo, Sanders and Oreopoulos 2010), 44,362 rows	fuzzy RDD	distance to the probation cutoff ranked first of four forcing candidates; cutoff at 0; 2SLS sign matches the published +0.233
Emergency-department visits (Anderson, Dobkin and Gross 2012), 161 monthly cells	fuzzy RDD	insurance-loss cutoffs at ages 19 and 23 among the top clusters; density negative control holds
Inpatient visits (same source), 73 cells	fuzzy RDD	age-23 cutoff on 73 aggregated cells
Bavarian council size (Egger and Koethenbuerger 2010), 42,005 rows	multi-cutoff RDD	four of five statutory population thresholds are the only top-50 clusters; the fifth ranks 54th
Proposition 99 (Abadie, Diamond and Hainmueller 2010), 39 states $\times$ 31 years	SuDDDS and SC	(California, 1989) recovered exactly by all scan methods; Bernoulli-null $p = 0.010$; donor weight 0.955 on the five published donors; ATT -19.5 packs; in-space placebo rank 3 of 39

7.2 Size of nominal inference in the field

Figure 1 and Table 7 report, for eight AI time series, the share of shifted placebo cutoffs at which the nominal HC1 test rejected at 5%. The first four are the flagship paper’s series [8]. The capex series is from its companion note [6]. The last three are the October 2026 runs of Section 7.3. The nominal test rejected at between 7% and 88% of non-event dates. On no series did it hold its size. The placebo-calibrated p reaches its floor on three series. They are the GPQA-Diamond score at o1-preview and two series at the Mythos announcement: pooled high and critical CVE publications, and fresh additions to the CISA Known Exploited Vulnerabilities catalogue. The series with the largest nominal z , the GPU-cluster stock at ChatGPT with $z = 8.95$, has a calibrated p of 0.62. A smooth super-exponential series bends everywhere.

7.3 Worked example: vulnerability disclosure and exploitation after Claude Mythos

Anthropic announced Claude Mythos and Project Glasswing on 7 April 2026, with eight founding partners that included several large vendors and the Linux Foundation. The question was whether vulnerability disclosure bent at that date and whether exploitation followed. The data are the CVE Program’s cvelistV5 bundle of 2 October 2026 [11] and the CISA Known Exploited Vulnerabilities catalogue of the same day [12]. The CVE records are aggregated to an organisation by month panel over Epoch AI’s 21 notable CVE numbering authorities.

Table 8 reports the kink at April 2026 in the log of monthly high and critical publications, with its robustness and placebo-date rows. The pooled slope change is 3.9 log points per year. It is the most extreme of 49 placebo dates. It holds when the Linux kernel authority, the founding partners,

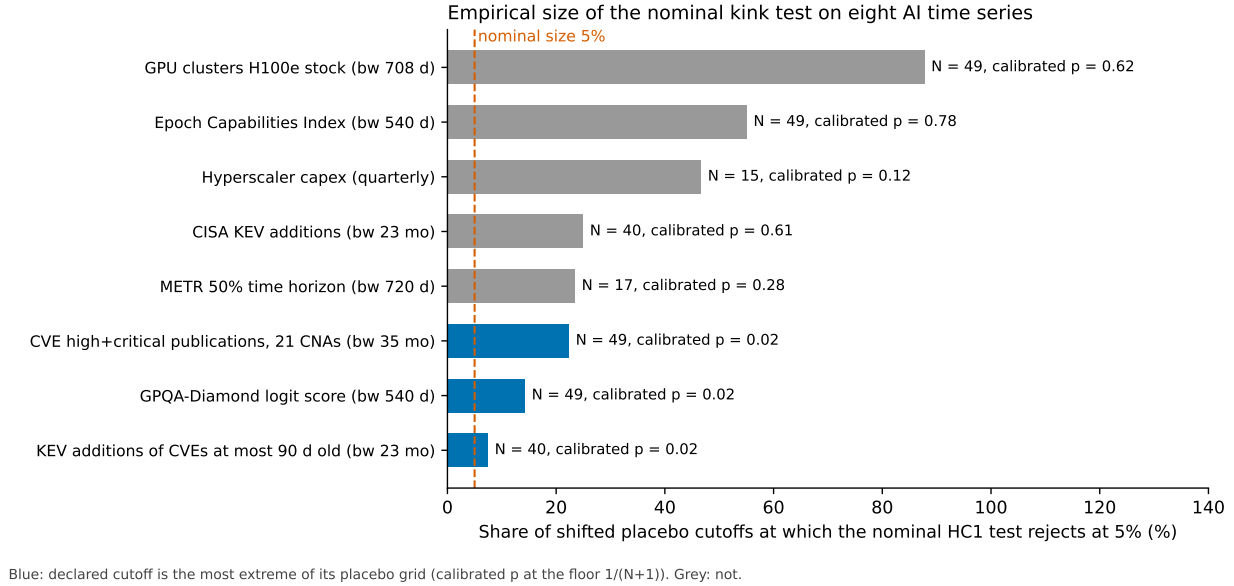


Figure 1: Empirical size of the nominal kink test at shifted placebo cutoffs. Blue bars mark series whose declared cutoff is the most extreme of its grid. Data: `figures/placebo_size.csv`.

Table 7: Placebo grids on eight AI time series. N is the number of evaluable placebo cutoffs; the rejection rate is the share at which the nominal HC1 test rejects at 5%.

Series	Event	N	Rejection	p_{cal}	Verdict
GPQA-Diamond logit score	o1-preview	49	14%	0.020	dated bend
METR 50% time horizon	o1-preview	17	24%	0.278	era bend
Epoch Capabilities Index	o1-preview	49	55%	0.780	null
GPU clusters, H100e stock	ChatGPT	49	88%	0.620	null
Hyperscaler capex, quarterly	ChatGPT	15	47%	0.125	inconclusive
CVE high and critical, 21 CNAs	Mythos	49	22%	0.020	dated bend
CISA KEV additions	April 2026	40	25%	0.610	null
KEV additions of CVEs at most 90 days old	April 2026	40	8%	0.024	dated bend

or the non-partners are removed in turn. Shifting the declared date to April 2025 or April 2024 gives calibrated p -values of 0.18 and 0.86. Blind SuDDDS scans, told nothing about the date, put the jump at April or May 2026 for all 21 authorities. A two-way fixed-effects contrast of founding partners against the other 13 authorities after April 2026 is 0.42 log points. Its cluster-robust p is 0.33 and its placebo-in- space p is 0.41 over 1,999 reassignments. The surge is broad-based; these data do not attribute it to the partners.

Exploitation did not keep pace (Table 9). Total monthly additions to the KEV catalogue show no bend at April 2026. Blind scans put the catalogue’s sharpest change in 2022, when it was being populated. Additions of CVEs published within the previous 90 days did bend, by 1.7 log points per year, at the floor of a 40-position grid. Because disclosures rose faster, the share of published CVEs that entered the catalogue within 30 days fell from 2.7 to 1.9 per thousand. The median lag from publication to catalogue entry fell from 7 to 4 days. The Mann–Whitney p is 0.12 on 189 and 149 entries, and the post-period lags are right-censored. A vendor-level contrast of the seven

Table 8: Kink at April 2026 in log monthly high and critical CVE publications, 35-month bandwidth. Slope changes are in log points per year.

Series	Slope change	z	N	Rejection	p_{cal}
21 notable CNAs	3.88	11.7	49	22%	0.02
without the Linux kernel CNA	3.96	11.7	49	41%	0.02
13 non-partner CNAs	5.12	7.7	49	45%	0.02
8 founding partners	3.23	9.1	49	51%	0.02
all CNAs	2.52	15.3	49	35%	0.02
21 notable CNAs, placebo date April 2025	0.72	2.4	49	27%	0.18
21 notable CNAs, placebo date April 2024	-0.16	-0.7	49	51%	0.86

Table 9: Kinks at April 2026 in log monthly KEV additions, 23-month bandwidth.

Series	Slope change	z	N	Rejection	p_{cal}
all additions	0.81	1.1	40	25%	0.61
additions of CVEs at most 90 days old	1.74	3.8	40	8%	0.02
additions within 30 days of publication	1.18	1.8	40	3%	0.12

partner vendors against 29 others after April 2026 is 0.14 log points. Its cluster-robust p is 0.42 and its placebo-in-space p is 0.19.

7.4 An honest miss

One collection note documents a failure of the level-break scan on a diffusion process [7]. A blind LoRD3 scan of $\log_{10}$ training tokens per parameter over calendar time was run on 549 language models. It returned a single significant discovery centred on 22 September 2022. The known event, the Chinchilla scaling result of 29 March 2022, ranks 158th of 547 candidates. The discovered break is six months late and points the wrong way. Fine-tuned releases enter the panel with token counts that cover only the fine-tuning run. Two rules follow. Diffusion-style treatments need a slope statistic, not only a level-break statistic. And fine-tune rows must be filtered from tokens-per-parameter panels before any scan.

8 Discussion and limitations

What the calibrated p establishes is narrow. It says that the declared date is more extreme than every shifted date on the same series under the same estimator. It does not say what caused the bend. For the CVE series, the April 2026 bend coincides with the Mythos announcement. These data cannot attribute it to Mythos, to Glasswing, or to any single cause. The partner contrast is null and the bend is present in every authority group. For the GPQA-Diamond series the bend coincides with the entry of reasoning models into the release stream, and composition is the mechanism. A dated bend is a dated bend.

The limitations are these. Calendar-time kinks rest on one untestable assumption: no other slope-changing event is co-located with the declared date. Placebo calibration tests exchangeability of dates, not that assumption. HAC covariances are a mitigation on short persistent windows, not a certificate, which is why they do not decide verdicts. Robust bias correction for the local-polynomial fits is not implemented, so bandwidth and polynomial sensitivity are part of every analysis. The did family estimates effects for the configuration’s single declared outcome; per-outcome effects

exist only for rdd. The sc family can pick a time-invariant group indicator as the treatment on some panels, which is an open issue. The plug-in Lasso assumes a noisy first stage and over-selects when assignment is a deterministic step function. Synthetic controls fit outcomes only, without covariate weights. The analyst backends order the scan and never touch a statistic. A language model can neither improve nor corrupt the numbers. It can only change which configuration is reported first. The sixteen companion notes written under 0.2.0 have not been re-run under these rules. A calendar-time result in them that rests on a seven-placebo grid has a floor of 0.125. It is not a 5% finding.

9 Reproducibility

natex is a Python package with core dependencies limited to numpy, scipy, pandas, scikit-learn, typer and pydantic. Every stochastic call takes one `numpy.random.Generator`; a run replays bitwise from its seed. The offline test suite collects 1,269 tests and runs on Python 3.11 to 3.14 in continuous integration. The 32 real-data backtests of Section 7.1 run against a local data directory that is never committed. The first four series of Section 7.2 are taken from the flagship paper’s case-study file. The fifth is from the capex note. The last three are from the archived October 2026 runs. The figure is drawn from the committed file `figures/placebo_size.csv` by `make_figures.py`. Source, method cards, the reconciled audit and the release notes for 0.3.0 are in the repository [10].

A Placebo grid construction

Given a running variable r , a declared cutoff c , a bandwidth h and a target of $N^* = 49$ positions:

1. Take the distinct values of r between its 10th and 90th percentiles.
2. Drop values with $|r - c| \leq h/4$.
3. If more than N^* remain, keep N^* evenly spaced in rank order.
4. At each remaining value re-estimate the kink with the headline bandwidth, kernel, degree, donut and covariance. Positions where a cell lacks residual degrees of freedom are dropped. N counts the evaluable remainder.
5. Report p_{cal} , N and the floor $1/(N + 1)$. Report the share of positions with nominal $p \leq 0.05$.

The minimum of 19 positions is the smallest N for which the floor is at most 0.05.

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